

Matrix inequalities and norms

# Thompson-type domination for the arithmetic symmetric modulus

**Difficulty:** challenging **Importance:** interesting to specialist**Status:** Lean verified

**Rating rationale:** The arithmetic modulus lacks the useful order structure of the quadratic modulus; a solution would advance a focused family of operator triangle estimates.

## Lean proof and verification evidence — 2026-09-12

The original MI-06 assertion with factor  $\sqrt{2}$  is false, with a complete Lean-verified counterexample. At the rational pair

$$A = \begin{pmatrix} 1 & 3/4 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ -3/4 & 0 & 0 \end{pmatrix},$$

all six genuine matrix moduli are proved exactly. For every complex unitary pair, a nonzero common orthogonal vector gives a left quadratic form at least  $(3/8)\|w\|_2^2$  and a right quadratic form at most  $(\sqrt{2}/4)\|w\|_2^2$ . The strict scalar gap follows from the kernel-checked LeanCert certificate  $2 < 9/4$ . The [proof at revision 43b3dc6](#) retains the full original complex-matrix and all-unitary target.

**Mathematical counterexample and informal proof:** Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Lean formalization:** George Stephanians, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA, with AI-agent assistance.

The six checked declarations in [Solution.lean](#) are:

- `NLA.MI06.modulus_eq_sqrt`: the genuine positive square-root modulus.
- `NLA.MI06.witness_moduli`: six exact moduli, their averages and rank-one decompositions.
- `NLA.MI06.two_vector_orthogonal`: a nonzero common orthogonal vector in complex dimension three.
- `NLA.MI06.witness_quadratic_bounds`: homogeneous bounds for every complex unitary pair.
- `NLA.MI06.counterexample`: the fixed pair defeats all unitary choices.
- `NLA.MI06.not_dominationConjecture`: negation of the complete original assertion.

Two independent agents reviewed the [frozen statements and completed proof](#). [Linux run 34711237623](#) matched all six exports with the sandboxed Comparator and replayed them through Lean’s default kernel. The [original artifacts and independent operational audit](#) retain all 117 source hashes, isolation checks and rejection controls. All 52 [transitive axiom reports](#) contain only *propext*, *Classical.choice* and *Quot.sound*. These are independent agent reviews, not external human peer review.

The proof pins [Lean 4.33.1](#), [LeanCert 621a43d](#) and [Mathlib 0df444a](#). See the [manifest](#), [dependency pins](#) and [numerical targets](#). On a documented [non-root Linux host](#), reproduce from the verified revision with:

```
tools/lean/bootstrap.sh /absolute/path/to/nla-lean-tools
tools/lean/selftest.sh /absolute/path/to/nla-lean-tools
tools/lean/verify.sh \
  matrix-inequalities-and-norms/MI-06/lean \
  /absolute/path/to/nla-lean-tools
```

The stronger no-finite-constant theorem in the original manuscript is outside these six formal exports. The original informal proof and its historical audit remain below.

**Resolution — 2026-09-11**

**Negative result by Matthew J. Colbrook**, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent proof review: PASS.**

No finite constant permits the proposed two-unitary Loewner-order domination for the arithmetic symmetric modulus, already in dimension three. A fixed rational example also refutes the proposed  $\sqrt{2}$  constant. This concerns matrix order, not a separate norm triangle inequality.

The exact target is resolved. The original statement and source evidence are retained below; its former difficulty rating is historical.

**Primary manuscript:** [complete proof PDF](#), [standalone TeX](#), Theorem 1.1 and its proof; [authorship and scope](#). The [independent review](#) checks the full original argument and records its hash. The draft was AI-assisted; the 2026-09-11 review was independent agent verification, without external human peer review or formal certification. The later Lean verification above covers the complete original factor- $\sqrt{2}$  target. [Submission record](#).

**Problem statement**

For  $X \in \mathbb{C}^{n \times n}$ , define  $|X| = (X^* X)^{1/2}$  and

$$S(X) = \frac{|X| + |X^*|}{2}.$$

For every  $n \geq 1$  and every  $X, Y \in \mathbb{C}^{n \times n}$ , do there exist unitary  $U, V \in \mathbb{C}^{n \times n}$  such that

$$S(X + Y) \leq \sqrt{2}(US(X)U^* + VS(Y)V^*)?$$

Here  $P \leq Q$  means  $Q - P$  is positive semidefinite. The conjectured universal factor  $\sqrt{2}$  is already known to be necessary.

**Why it matters**

Symmetrizing the left and right polar factors removes their directional preference. The question asks for a precise matrix-order bound for sums, which would yield singular-value and norm consequences for these positive representatives.

## References

1. T. Zhang, *Operator symmetric moduli and sharp triangle inequalities*, arXiv:2603.01046v1 (1 March 2026), §1.2, Conjecture 1.6. [Primary text](#).
2. J.-C. Bourin and E.-Y. Lee, *Some hybrid matrix triangle inequalities*, arXiv:2606.29188v1 (28 June 2026), introduction, Theorems 1.1 and 1.3. [Primary text](#).

## Status check — 2026-09-10

Both latest arXiv records remain v1. The June paper supplies related estimates but no proof of the displayed two-unitary inequality. Searches included *Zhang Conjecture 1.6 symmetric moduli*, *arithmetic symmetric modulus Thompson conjecture 2026*, and the two titles with *proof*. No resolution was located. The analogous theorem for the quadratic symmetric modulus is proved and must not be confused with this conjecture; a formula labeled Conjecture 3.13 in arXiv:2606.15624 appears to misidentify the modulus, so the draft follows Zhang’s original statement.

**Audit update (2026-09-10):** Rechecked Zhang Conjecture 1.6 and Bourin–Lee Theorem 1.3, with searches for later symmetric-modulus resolutions. Submajorization is weaker than the required two-unitary positive-semidefinite domination. This is a bounded literature check, not a proof that no solution exists.